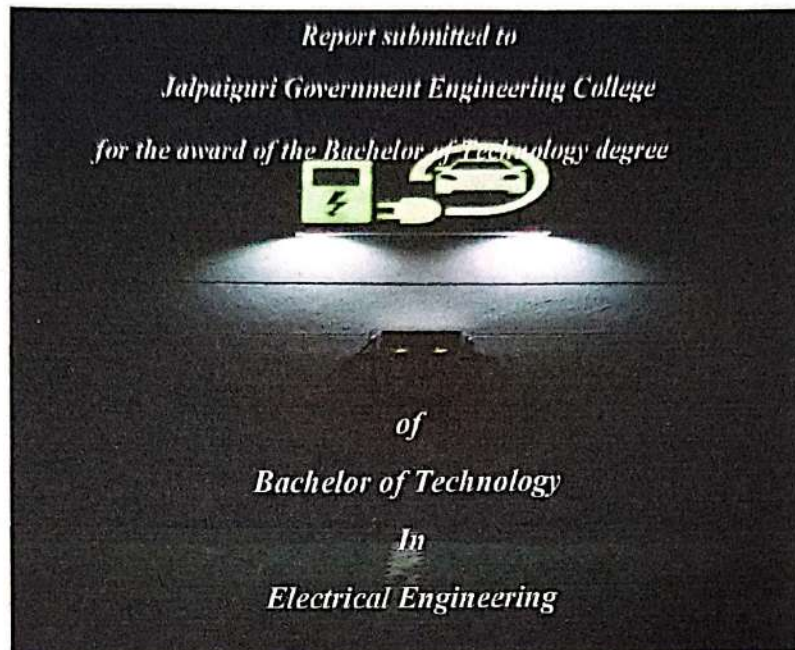


Title of the Project-MPPT Based Charging Circuit of Electric Vehicle



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JUNE, 2023

DECLARATION

We certify that

- a. the work contained in this report is original and has been done by me under the guidance of my supervisor(s).
- b. the work has not been submitted to any other Institute for any degree or diploma.
- c. We have followed the guidelines provided by the Institute in preparing the report.
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ABSTRACT

As the world transitions towards clean and sustainable energy solutions, the integration of solar photovoltaic (PV) systems with electric vehicle (EV) charging infrastructure has gained significant attention. By harnessing the power of the sun, solar PV charging of EV batteries offers a promising avenue for reducing greenhouse gas emissions and promoting renewable energy adoption. To optimize the charging efficiency, maximize energy transfer, and enhance overall system performance, MPPT (Maximum Power Point Tracking) technology is employed in these systems.

MPPT, a well-established technique in solar PV applications, ensures that solar panels operate at their maximum power point (MPP) under varying environmental conditions. When applied to EV charging, MPPT-based systems dynamically adjust the operating parameters of the solar panels to extract the highest possible power output. This allows for efficient energy conversion and optimal charging rates for EV batteries.

In an MPPT-based solar PV charging system for EVs, the integration of MPPT algorithms into the charging controller enables continuous monitoring and tracking of the MPP. By precisely controlling the voltage and current supplied to the batteries, the system maximizes power transfer and enhances the charging efficiency. The MPPT algorithm continually adjusts the operating point of the solar panels, adapting to changing solar irradiation, temperature, and shading conditions. This ensures that the system harvests the maximum available solar energy and delivers it to the EV batteries effectively.

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❖ INTRODUCTION: -

MPPT stands for Maximum Power Point Tracking. It is a technique used in solar photovoltaic (PV) systems to optimize the power output from solar panels. The goal of MPPT is to ensure that the solar panels operate at their maximum power point (MPP) under varying environmental conditions.

Solar panels have a non-linear relationship between their output voltage and current. The MPP is the specific combination of voltage and current at which the panels produce the maximum power. However, this point varies depending on factors such as solar irradiation, temperature, and shading.

MPPT algorithms continuously monitor and adjust the operating point of the solar panels to find and maintain the MPP. The algorithms employ various techniques to track the MPP, such as perturb-and-observe, incremental conductance, or hill-climbing algorithms. These techniques incrementally adjust the operating voltage of the panels to determine the point at which they produce the maximum power.

By tracking the MPP and adjusting the operating parameters, MPPT ensures that the solar panels operate at their highest efficiency, maximizing the power output and energy harvest. This allows for optimal utilization of available solar energy and improves the overall performance of the solar PV system.

MPPT is commonly used in grid-tied and off-grid solar PV systems, as well as in solar-powered applications such as solar-powered water pumping, solar-powered street lighting, and solar-powered charging systems for batteries, including EV (electric vehicle) batteries.

In summary, MPPT is a technique used in solar PV systems to track and maintain the maximum power point of the solar panels. By optimizing the power output, MPPT improves the energy harvest, increases system efficiency, and enables effective utilization of solar energy.

Level 2 Charging: 240-Volt or 240-Volt

Connectors Used: J1772, Tesla

Charging Speed: 12 to 19 miles per hour

Location: Home, Workplace, etc.

Level 2 charging is the most commonly used level for daily EV charging. Level 2 charging equipment is available at home, at work, at public charging stations, as well as at some commercial locations. Level 2 charging can provide a charging rate of 12 to 19 miles of range per hour, depending on the power output of the Level 2 charger and the vehicle's maximum charging rate. Most EV owners choose to use Level 2 charging equipment at home, as it allows them to charge their vehicle overnight, even if they plug in with a nearly empty battery.

Level 2 charging can deliver up to 20 miles of range per hour, but that requires a 240-volt Level 2 charger. Level 2 charging can also deliver up to 10 miles of range per hour, but that requires a 120-volt Level 2 charger. Level 2 charging can also deliver up to 5 miles of range per hour, but that requires a 120-volt Level 2 charger. Level 2 charging can also deliver up to 2 miles of range per hour, but that requires a 120-volt Level 2 charger.

❖ Different Charging System of EV:-

There are three levels of EV charging: Level 1, Level 2, and Level 3. Level 3 is broken into DC Fast Charging and (Tesla) Supercharging. The higher the level of charging, the faster the charging process, as more power is delivered to the vehicle. It's important to note that different EVs charge at different speeds on each level, because each EV can accept different levels of power from the EVSE, industry-speak for electric vehicle supply equipment, the charger.

When an electric vehicle is plugged in, there's a communication process before the charger is energized. Basically, the car asks the charger how much power it can deliver, and then the car calls for the maximum amount of power that the station can deliver and the vehicle can accept.

The car always determines how much power it accepts, so there's no need to worry about plugging into a charging station that can deliver more power than your EV can handle. The car will not allow the charger to deliver too much power.

Level 1 Charging: 120-Volt

Connectors Used: J1772, Tesla

Charging Speed: 3 to 5 Miles Per Hour

Locations: Home, Workplace & Public

Level 1 charging uses a common 120-volt household outlet. Every electric vehicle or plug-in hybrid can be charged on Level 1 by plugging the charging equipment into a regular wall outlet. Level 1 is the slowest way to charge an EV. It adds between 3 and 5 miles of range per hour.

Level 1 charging works well for plug-in hybrid electric vehicles (PHEVs) because they have smaller batteries, currently less than 25 kWh. Since EVs have much larger batteries, Level 1 charging is too slow for most daily charging, unless the vehicle isn't needed to drive very far on a daily basis. Most BEV owners find that Level 2 charging better suits their daily charging needs.

Level 2 Charging: 208-Volt to 240-Volt

Connectors Used: J1772, Tesla

Charging Speed: 12 to 80 Miles Per Hour

Locations: Home, Workplace & Public

Level 2 charging is the most commonly used level for daily EV charging. Level 2 charging equipment can be installed at home, at the workplace, as well as in public locations like shopping plazas, train stations and other destinations. Level 2 charging can replenish between 12 and 80 miles of range per hour, depending on the power output of the Level 2 charger, and the vehicle's maximum charge rate. Most BEV owners choose to install Level 2 charging equipment at their residence, because it charges the vehicle up to 10 times faster than Level 1 charging. Charging from a Level 2 source usually means the vehicle will be completely charged overnight, even if you plugged with a nearly empty battery.

Level 2 chargers can deliver up to 80 amps of power. But that requires a 100-amp 208-240V dedicated circuit and a heavy, costly supply line from the breaker box. Most owners will be well served choosing a 40-amp charger that can deliver 9.6 kW to the EV. A 48-amp charger can charge slightly faster at 11.5 kW, but requires a heavier gauge wire and the charger must

be hardwired to comply with the NEC code. Therefore, 48-amp chargers can cost significantly more than a 40-amp unit and offer only marginally faster charging.

Level 3 Charging: 400-Volt to 900-Volt (DC Fast Charge & Supercharging)

Connectors Used: Combined Charging System (Combo), CHAdeMO & Tesla

Charging Speed: 3 to 20 Miles Per Minute

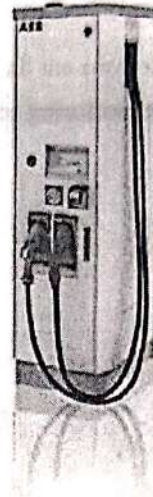
Locations: Public

Level 3 charging is the fastest type of charging available and can recharge an EV at a rate of 3 to 20 miles of range per minute. Unlike Level 1 and Level 2 charging that uses alternating current (AC), Level 3 charging uses direct current (DC). The voltage is also much higher than Level 1 & 2 charging, which is why you don't see level 3 chargers installed at home. Very few residential locations have the high-voltage supply that is required for level 3 charging.

Additionally, DC Fast Chargers cost tens of thousands of dollars. So even if your residence has 400-volt electricity service, the cost to install the charger would most likely cost more than your EV. Tesla calls their Level 3 chargers Superchargers; others are called DC Fast Chargers. Current Nissan EVs use a third specification.

1. DC Fast Charging Explained

AC charging is the simplest kind of charging to find – outlets are everywhere and almost all EV chargers you encounter at homes, shopping plazas, and workplaces are Level 2 AC chargers. An AC charger provides power to the on-board charger of the vehicle, converting that AC power to DC in order to enter the battery. The acceptance rate of the on-board charger varies by brand but is limited for reasons of cost, space and weight. This means that depending on your vehicle it can take anywhere from four or five hours to over twelve hours to fully charge at Level 2.



DC Fast Charging bypasses all of the limitations of the on-board charger and required conversion, instead providing DC power directly to the battery, charging speed has the potential to be greatly increased. Charging times are dependent on the battery size and the output of the dispenser, and other factors, but many vehicles are capable of getting an 80% charge in about or under an hour using most currently available DC fast chargers.

DC fast charging is essential for high mileage/long distance driving and large fleets. The quick turnaround enables drivers to recharge during their day or on a small break as opposed to being plugged in overnight, or for many hours, for a full charge.

Older vehicles had limitations that only allowed them to charge at 50kW on DC units (if they were able to at all) but newer vehicles are now coming out that can accept up to 270kW. Because battery size has increased significantly since the first EVs hit the market, DC chargers have been getting progressively higher outputs to match – with some now being capable of up to 350kW.

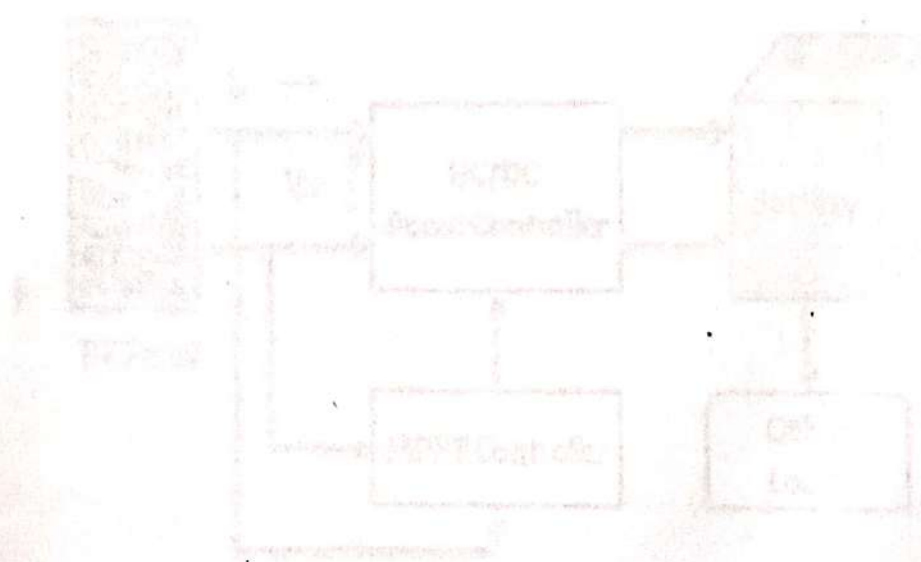
Currently, in North America there are three types of DC fast charging: CHAdeMO, Combined Charging System (CCS) and Tesla Supercharger.

All major DC charger manufacturers offer multi-standard units that offer the ability to charge via CCS or CHAdeMO from the same unit. The Tesla Supercharger can only service Tesla vehicles, however Tesla vehicles are capable of using other chargers, specifically CHAdeMO for DC fast charging, via an adapter.

COMBINED CHARGING SYSTEM (CCS)

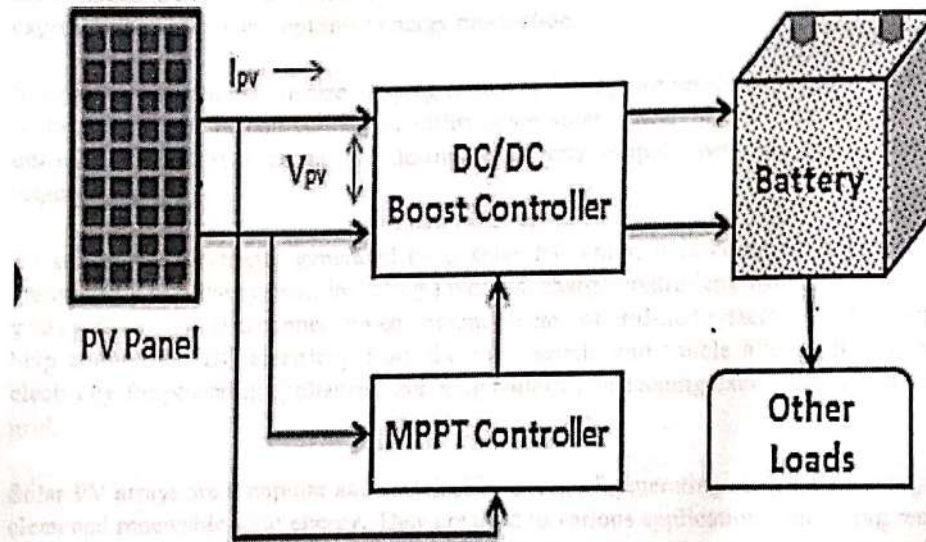
The Combined Charging System (CCS) is based on open and universal standards for electric vehicles. The CCS combines single-phase AC, three-phase AC and DC high-speed charging in both Europe and the US – all in a single, easy to use system.

The CCS includes the connector and inlet combination as well as all the control functions. It also manages communications between the electric vehicle and the infrastructure. As a result, it provides a solution to all charging requirements.



MPPT based Charging System

Block Diagram-



❖ Components:

- **Solar PV Array:**

A solar PV array, also known as a solar panel array or solar array, refers to a collection or grouping of multiple solar photovoltaic (PV) panels that are interconnected and working together to generate electricity from sunlight.

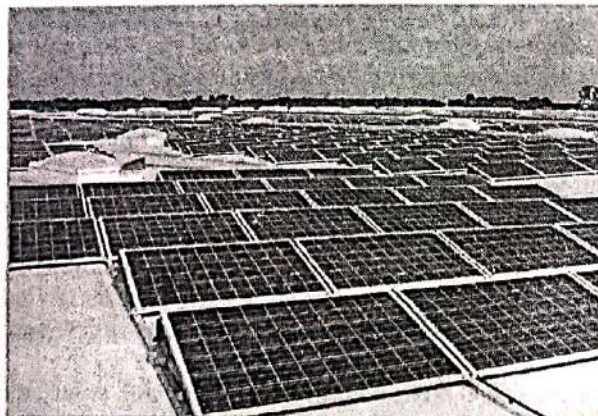
Each individual solar PV panel within the array consists of multiple solar cells that convert sunlight into direct current (DC) electricity through the photovoltaic effect. These cells are typically made of semiconductor materials, such as silicon, which generate an electric current when exposed to sunlight.

When multiple solar panels are connected in an array, their combined power output is increased, allowing for the generation of larger amounts of electricity. The panels in the array are arranged in a specific configuration, often mounted on rooftops or open spaces, to maximize exposure to sunlight and optimize energy production.

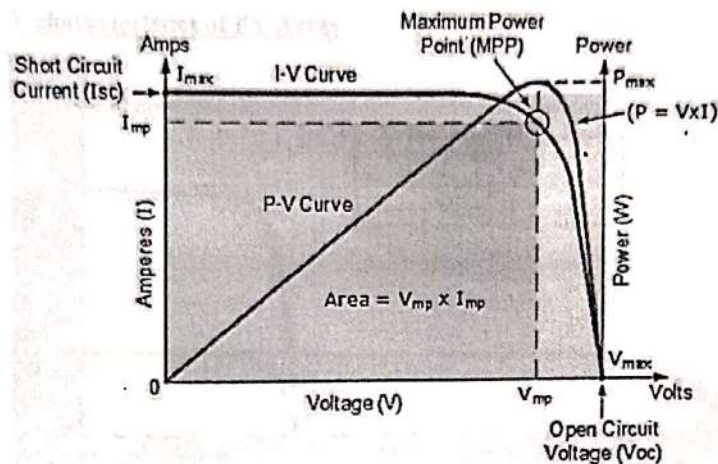
Solar PV arrays can vary in size, ranging from small arrays for residential applications to large-scale arrays used in commercial or utility-scale solar power plants. The size of the array depends on factors such as the desired electricity output, available space, and energy requirements.

To utilize the electricity generated by a solar PV array, it is common to integrate other components into the system, including inverters, charge controllers, batteries (in case of off-grid systems), and grid connection equipment (in case of grid-tied systems). These components help convert the DC electricity from the solar panels into usable alternating current (AC) electricity for powering appliances, charging batteries, or feeding excess energy back into the grid.

Solar PV arrays are a popular and sustainable means of generating electricity as they harness clean and renewable solar energy. They are used in various applications, including residential, commercial, industrial, and utility-scale installations, contributing to reducing dependence on fossil fuels and mitigating greenhouse gas emissions.



- **Solar Cell I-V & P-V Characteristic Curve:**



The above graph shows the current-voltage (I-V) characteristics of a typical silicon PV cell operating under normal conditions. The power delivered by a single solar cell or panel is the product of its output current and voltage ($I \times V$). If the multiplication is done, point for point, for all voltages from short-circuit to open-circuit conditions, the power curve above is obtained for a given radiation level.

With the solar cell open-circuited, that is not connected to any load, the current will be at its minimum (zero) and the voltage across the cell is at its maximum, known as the solar cells, **open circuit voltage**, or V_{oc} . At the other extreme, when the solar cell is short circuited, that is the positive and negative leads connected, the voltage across the cell is at its minimum (zero) but the current flowing out of the cell reaches its maximum, known as the solar cells **short circuit current**, or I_{sc} .

Then the span of the solar cell I-V characteristics curve ranges from the short circuit current (I_{sc}) at zero output volts, to zero current at the full open circuit voltage (V_{oc}). In other words, the maximum voltage available from a cell is at open circuit, and the maximum current at closed circuit. Of course, neither of these two conditions generates any electrical power, but there must be a point somewhere in between where the solar cell generates maximum power.

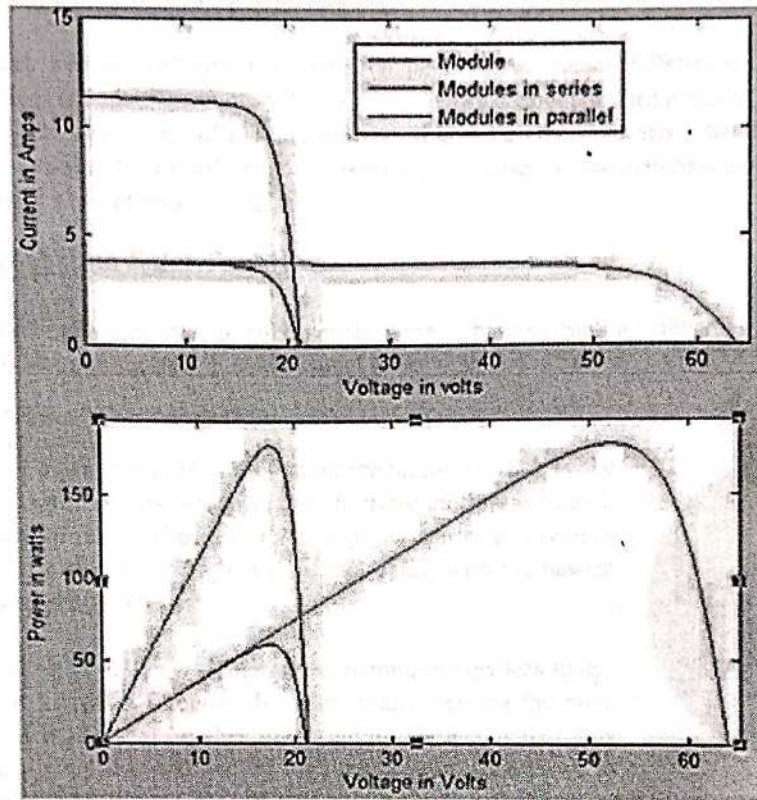
However, there is one combination of current and voltage for which the power reaches its maximum value, at I_{mp} and V_{mp} . In other words, the point at which the cell generates maximum electrical power, and this is shown at the top right area of the green rectangle. This is the "maximum power point" or **MPP**. Therefore, the ideal operation of a photovoltaic cell (or panel) is defined to be at the maximum power point.

The maximum power point (MPP) of a solar cell is positioned near the bend in the I-V characteristics curve. The corresponding values of V_{mp} and I_{mp} can be estimated from the open circuit voltage and the short circuit current: $V_{mp} \cong (0.8-0.90)V_{oc}$ and $I_{mp} \cong (0.85-0.95)I_{sc}$. Since solar cell output voltage and current both depend on temperature, the actual output power will vary with changes in ambient temperature.

Thus far we have looked at **Solar Cell I-V Characteristic Curve** for a single solar cell or panel. But a photovoltaic array is made up of smaller PV panels interconnected together. Then

the I-V curve of a PV array is just a scaled-up version of the single solar cell I-V characteristic curve as shown.

I-V and P-V characteristics of PV Array



❖ Different Types of Solar PV array:-

Different types of solar panels serve different needs and purposes. Given that sunlight can be used differently whether on Earth or in space points to the fact that location, itself, is a significant factor when it comes to choosing one of the types of solar panels over another.

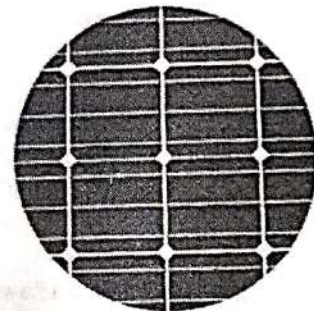
Distinguishing between different types of solar panels often means differentiating between single-junctions and multi-junctions solar panels—or first, second, or third generations. Single-junction and multi-junctions differ in the number of layers on the solar panel that will observe the sunlight, whereas the classification by generation focusses on the materials and efficiency of the different types of solar panels.

• 1st Generation Solar Panels:-

These are the traditional types of solar panels made of monocrystalline silicon or polysilicon and are most commonly used in conventional surroundings.

Monocrystalline Solar Panels (Mono-SI)

This type of solar panels (made of monocrystalline silicon) is the purest one. You can easily recognise them from the uniform dark look and the rounded edges. The silicon's high purity causes this type of solar panel has one of the highest efficiency rates, with the newest ones reaching above 20%.

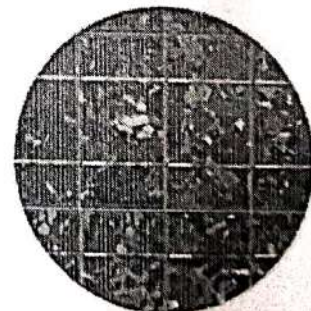


Monocrystalline panels have a high-power output, occupy less space, and last the longest. Of course, that also means they are the most expensive of the bunch. Another advantage to consider is that they tend to be slightly less affected by high temperatures compared to polycrystalline panels.

Polycrystalline Solar Panels (Poly-SI)

You can quickly distinguish these panels because this type of solar panels has squares, its angles are not cut, and it has a blue, speckled look. They are made by melting raw silicon, which is a faster and cheaper process than that used for monocrystalline panels.

This leads to a lower final price but also lower efficiency (around 15%), lower space efficiency, and a shorter lifespan since they are affected by hot temperatures to a greater degree. However, the differences between mono- and polycrystalline types of solar panels are not so significant and the choice will strongly depend on your specific situation. The first option offers a slightly higher space efficiency at a slightly higher price but power outputs are basically the same.

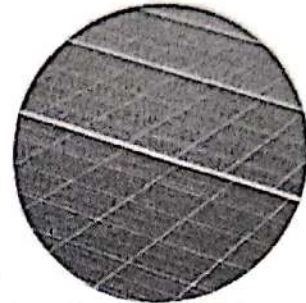


• 2nd Generation Solar Panels:-

These cells are different types of thin film solar cells and are mainly used for photovoltaic power stations, integrated in buildings or smaller solar systems.

Thin-Film Solar Cells (TFSC)

If you are looking for a less expensive option, you might want to look into thin-film. Thin-film solar panels are manufactured by placing one or more films of photovoltaic material (such as silicon, cadmium or copper) onto a substrate. These types of solar panels are the easiest to produce and economics of scale make them cheaper than the alternatives due to less material being needed for its production.



They are also flexible—which opens a lot of opportunities for alternative applications—and is less affected by high temperatures. The main issue is that they take up a lot of space, generally making them unsuitable for residential installations. Moreover, they carry the shortest warranties because their lifespan is shorter than the mono- and polycrystalline types of solar panels. However, they can be a good option to choose among the different types of solar panels where a lot of space is available.

Amorphous Silicon Solar Cell (A-Si)

Have you ever used a solar powered pocket calculator? Yes? Then you have definitely seen these types of solar panels before. The amorphous silicon solar cell is among the different types of solar panels, the one that is used mainly in such pocket calculators. This type of solar panel uses a triple layered technology, which is the best of the thin film variety.

Just to give a brief impression of what “thin” means, in this case, we’re talking about a thickness of 1 micrometre (one millionth of a metre). With only 7% efficiency rate, these cells are less effective than crystalline silicon ones—that have an efficiency rate of circa 18%—but the advantage is the fact that the A-Si-Cells are relatively low in cost.

- **3rd Generation Solar Panels:-**

3rd generation solar panels include a variety of thin film technologies but most of them are still in the research or development phase. Some of them generate electricity by using organic materials, others use inorganic substances (CdTe for instance).

Biohybrid Solar Cell

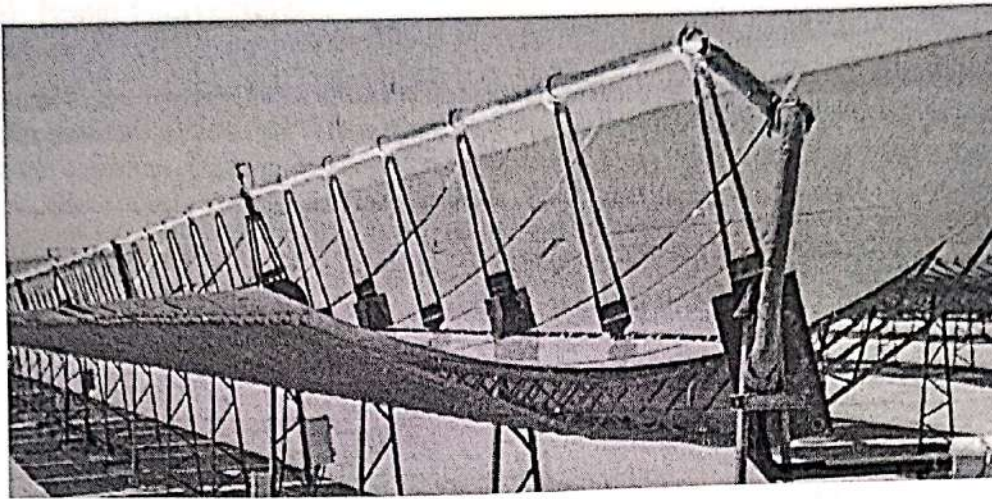
The Biohybrid solar cell is one of the types of solar panels, that is still in the research phase. It has been discovered by an expert team at Vanderbilt University. The idea behind the new technology is to take advantage of the photosystem 1 and thus emulate the natural process of photosynthesis. In case you want to learn more about how the biohybrid solar cell works in detail, read more about it in the American Journal of Optics and Photonics. It explains more detailed how these cells work. Many of the materials being used in this cell are similar to the traditional methods, but only by combining the multiple layers of photosystem 1, the conversion from chemical to electrical energy becomes much more effective (up to 1000 times more efficient than 1st generation types of solar panels).

Cadmium Telluride Solar Cell (CdTe)

Among the collection of different types of solar panels, this photovoltaic technique uses Cadmium Telluride, which enables the production of solar cells at relatively low cost and thus a shorter payback time (less than a year). Of all solar energy technologies, this is the one requiring the least amount of water for production. Keeping the short energy payback time in mind, CdTe solar cells will keep your carbon footprint as low as possible. The only disadvantage of using Cadmium Telluride is its characteristic of being toxic, if ingested or inhaled. In Europe especially, this is one of the greatest barriers to overcome, as many people are very concerned about using the technology behind this type of solar panel.

Concentrated PV Cell (CVP and HCVP)

Concentrated PV cells generate electrical energy just as conventional photovoltaic systems do. Those multi-junction types of solar panels have an efficiency rate up to 41%, which, among all photovoltaic systems, is the highest so far.



The name of such CVP cells is related to what makes them so efficient, compared to other types of solar panels: curved mirror surfaces, lenses and sometimes even cooling systems are used to bundle the sun rays and thus increase their efficiency.

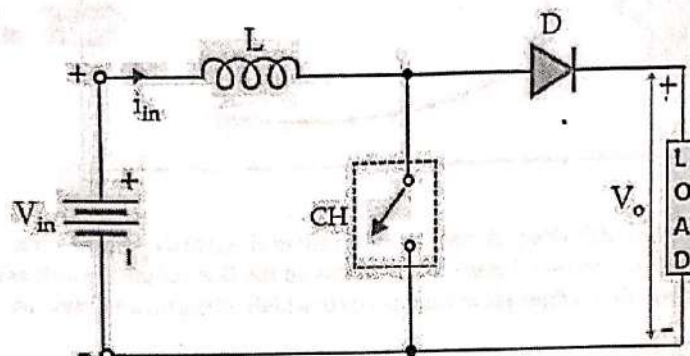
By this means, CVP cells have become one of the most efficient solar panels, with a high performance and efficiency rate of up to 41%. What remains is the fact, that such CVP solar panels can only be as efficient if they face the sun in a perfect angle. In order to reach such high efficiency rates, a solar tracker inside the solar panel is responsible for following the sun.

❖ Boost Converter:-

Boost Converters sometimes, also known as step-up choppers are the type of chopper circuits that provides such an output voltage that is more than the supplied input voltage. In the case of boost converters, the dc-to-dc conversion takes place in a way that the circuit provides a high magnitude of output voltage than the magnitude of the supply input.

Working Principle

The figure given below is the circuit representation of the boost converter:

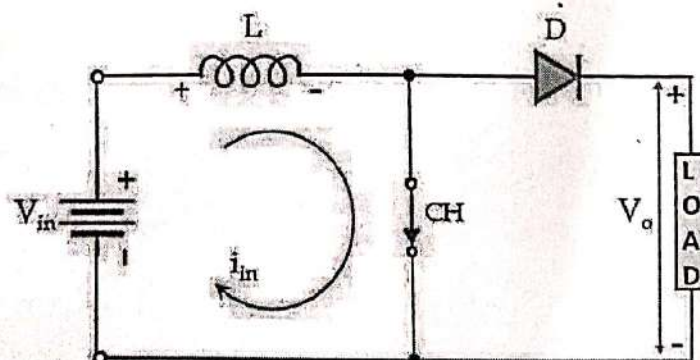


Elementary Circuit of Boost Converter

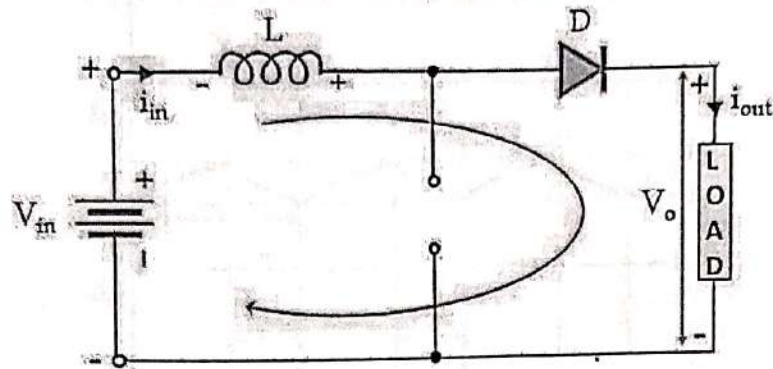
The circuit here is an elementary form of step-up chopper which necessarily requires a large inductor L in series connection with the voltage source. The whole circuit arrangement operates in a way that it helps in maintaining a regulated dc signal at the output.

Let us understand how the given circuit operates in order to provide an increased dc signal at the load.

Initially, when the chopper CH is in on state, then in the presence of supply dc input current begins to flow through the closed path of the circuit i.e., passing through the inductor as shown in the figure below.



Here, the polarity of the inductor will be according to the direction of the flow of current. In this case, the diode in the configuration is in reverse biased condition and so current will not be allowed to flow through that part of the circuit during on state of the chopper. Resultantly, the voltage across the chopper will appear across the load.



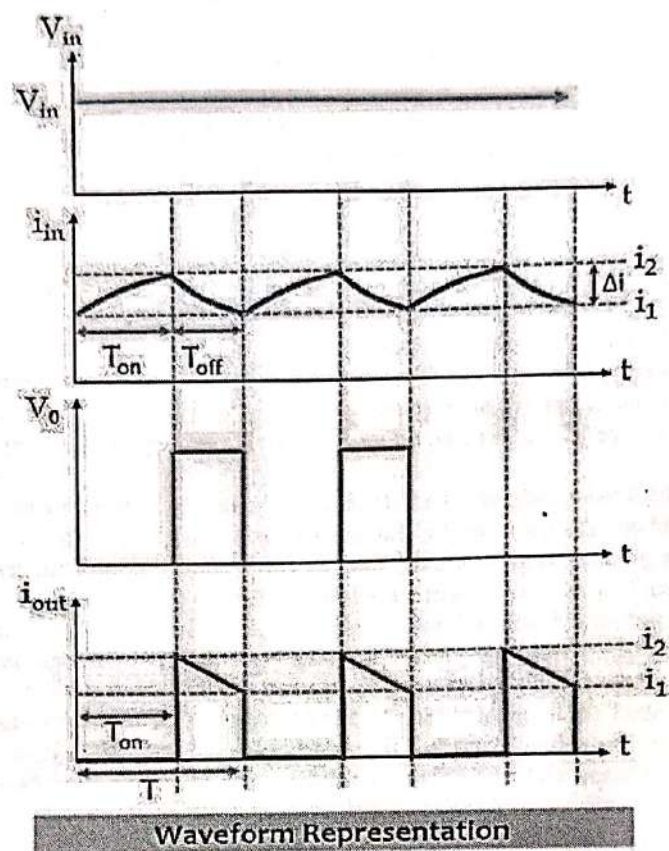
Furthermore, at the instant when CH is in the off state, then the part of the circuit through which the current was flowing earlier will not be active in this case. However, as the inductor stores the energy in the form of a magnetic field and so the current through it will not die out instantly.

Also, we know according to Lenz's law a reverse current will be induced that will oppose the cause which has produced it. And so, due to the induced current, the polarity of the inductor will get reversed. This reverse polarity of the inductor forward biases the diode present in the circuit. This provides the path for the current through the diode that flows through the load during the off state of the chopper i.e., T_{off} . However, we must note here that the current through the inductor is of decreasing nature and will die out after a point in time.

Thus, the total voltage across the load will be given as:

$$V_{out} = V_{in} + V_L$$

The waveform representation of the boost converter shown below:



❖ MPPT (Maximum Power Point Tracking):-

MPPT stands for Maximum Power Point Tracking, a technique to regulate the charge of your battery bank. The function of an MPPT charge controller is analogous to the transmission in a car. When the transmission is in the wrong gear, the wheels do not receive maximum power. That's because the engine is running either slower or faster than its ideal speed range. The purpose of the transmission is to couple the engine to the wheels, in a way that lets the engine run in a favorable speed range in spite of varying acceleration and terrain.

Let's compare a PV module to a car engine. Its voltage is analogous to engine speed. Its ideal voltage is that at which it can put out maximum power. This is called its maximum power point. (It's also called peak power voltage, abbreviated V_{pp}). V_{pp} varies with sunlight intensity and with solar cell temperature. The voltage of the battery is analogous to the speed of the car's wheels. It varies with battery state of charge, and with the loads on the system (any appliances and lights that may be on). For a 12V system, it varies from about 11 to 14.5V.

In order to charge a battery (increase its voltage), the PV module must apply a voltage that is higher than that of the battery. If the PV module's V_{pp} is just slightly below the battery voltage, then the current drops nearly to zero (like an engine turning slower than the wheels). So, to play it safe, typical PV modules are made with a V_{pp} of around 17V when measured at a cell temperature of 25°C. They do that because it will drop to around 15V on a very hot day. However, on a very cold day, it can rise to 18V!

What happens when the V_{pp} is much higher than the voltage of the battery? The module voltage is dragged down to a lower-than-ideal voltage. Traditional charge controllers transfer the PV current directly to the battery, giving you NO benefit from this added potential.

Now, let's make one more analogy. The car's transmission varies the ratio between speed and torque. At low gear, the speed of the wheels is reduced and the torque is increased, right? Likewise, the MPPT varies the ratio between the voltage and current delivered to the battery, in order to deliver maximum power. If there is excess voltage available from the PV, then it converts that to additional current to the battery. Furthermore, it is like an automatic transmission. As the V_{pp} of the PV array varies with temperature and other conditions, it "tracks" this variance and adjusts the ratio accordingly. Thus, it is called a Maximum Power Point Tracker.

❖ How MPPT-based solar PV charging of EV batteries works?

1. **Solar PV Array:** The system consists of solar panels that convert sunlight into electrical energy. The panels generate direct current (DC) electricity, which needs to be converted and regulated for charging the EV battery.
2. **MPPT Charge Controller:** An MPPT charge controller is used to regulate the charging process and maximize the power output from the solar panels. It constantly monitors the solar panel's voltage and current and adjusts the operating parameters to operate the panels at their maximum power point.
3. **Maximum Power Point Tracking:** The MPPT algorithm employed by the charge controller adjusts the operating conditions of the solar panels to ensure they are operating at their maximum power point. The maximum power point is the voltage and current combination that results in the highest power output from the solar panels. The MPPT controller tracks changes in sunlight intensity and adjusts the panel voltage to maintain maximum power output.
4. **DC-DC Conversion:** The output of the MPPT charge controller is DC electricity, which needs to be converted to the appropriate voltage and current levels for charging the EV battery. A DC-DC converter is used for this purpose. It steps up or steps down the voltage to match the charging requirements of the EV battery.
5. **Battery Charging:** The DC-DC converter delivers the regulated DC electricity to the EV battery for charging. The charging process is controlled to ensure the battery is charged safely and efficiently.

❖ PWM (Pulse Width Modulation):-

PWM in the context of solar refers to Pulse Width Modulation, a technique widely used in solar power systems for efficient battery charging. In solar energy systems, photovoltaic panels generate direct current (DC) electricity from sunlight. However, the amount of energy produced by these panels can fluctuate due to factors like weather conditions or shading. Without proper regulation, these fluctuations can lead to inefficient charging or even damage to the batteries.

To address this issue, PWM charge controllers are employed. These controllers manage the charging process by rapidly switching the connection between the solar panels and the batteries on and off. By doing so, they create a series of short pulses of current, controlling the flow of energy to the batteries.

The principle behind PWM involves adjusting the width of these pulses based on the battery's state of charge. When the batteries are low on charge, wider pulses are generated, allowing a larger current to flow and facilitating faster charging. As the batteries approach their full capacity, the pulses become narrower, reducing the charging current to prevent overcharging.

By modulating the pulse width in this way, PWM charge controllers maintain a constant voltage during the charging process, protecting the batteries from excessive overcharging and extending their lifespan. This technique optimizes the charging current and ensures that the batteries receive the appropriate amount of energy at all times.

It's important to note that while PWM charge controllers are commonly used, there are alternative technologies available as well, such as Maximum Power Point Tracking (MPPT) controllers. MPPT controllers are particularly efficient in scenarios where the solar panels' output voltage significantly exceeds the battery voltage. Nonetheless, PWM charge controllers remain a reliable and cost-effective option for many solar power installations.

❖ Difference between MPPT based charging and PWM charging:-

MPPT (Maximum Power Point Tracking) and PWM (Pulse Width Modulation) are two different methods used for charging batteries in solar PV systems. Here are the key differences between MPPT-based charging and PWM charging:

1. **Charging Efficiency:** MPPT-based charging is more efficient compared to PWM charging. MPPT algorithms continuously track and adjust the operating point of the solar panels to maximize power transfer, ensuring that the panels operate at their maximum power point (MPP). This optimization results in higher charging efficiency, as more solar energy is harvested and utilized for charging the batteries. PWM charging, on the other hand, regulates the charging current by switching the solar panel output between fully on and fully off states, resulting in lower overall efficiency.
2. **Energy Harvesting:** MPPT charging systems can adapt to varying solar conditions, such as changes in irradiation, temperature, and shading. The algorithms dynamically adjust the operating parameters of the solar panels to extract the maximum available power. This enables the system to harvest more energy from the solar panels, even in less-than-ideal conditions. PWM charging does not have this adaptive capability, so it may not fully utilize the available solar energy.
3. **Voltage Conversion:** MPPT charging systems can step up or step down the voltage from the solar panels to match the battery voltage requirements. This allows for efficient power conversion and optimal charging rates. In PWM charging, the solar panel voltage is regulated by simply switching the panel output on and off. While this method can control the charging current, it does not provide voltage conversion capabilities.
4. **Compatibility:** MPPT charging systems are compatible with a wide range of battery types and sizes. The charging current can be adjusted to match the specific charging characteristics of the battery, optimizing the charging process and extending battery life. PWM charging is typically used for charging lead-acid batteries, where the charging voltage is regulated by adjusting the duty cycle of the PWM signal. It may not be as suitable for charging other types of batteries, such as lithium-ion batteries, which may require more precise voltage and current control.
5. **Cost and Complexity:** MPPT charging systems are generally more expensive and complex compared to PWM charging systems. MPPT chargers require additional electronic components, such as an MPPT controller, to track and adjust the operating point of the solar panels. This complexity adds to the overall cost of the system. PWM charging systems are simpler and more cost-effective, as they typically consist of a charge controller that regulates the charging current using PWM techniques.

In summary, MPPT-based charging offers higher charging efficiency, improved energy harvesting, voltage conversion capabilities, compatibility with various battery types, and adaptability to changing solar conditions. PWM charging, on the other hand, is simpler and more cost-effective, but it may not provide the same level of efficiency and adaptability as MPPT charging. The choice between MPPT and PWM depends on factors such as system requirements, battery type, and budget considerations.

❖ Advantages of MPPT:-

1. **Higher Efficiency:** MPPT maximizes the power output from solar panels by continuously tracking and adjusting the operating point to match the maximum power point (MPP) under varying environmental conditions. This ensures that the solar PV system operates at its highest efficiency, resulting in better energy conversion and increased charging efficiency. Higher efficiency means more energy is harvested from the solar panels, translating to faster and more effective charging of EV batteries.
2. **Improved Energy Harvest:** MPPT enables the solar PV system to adapt to changes in solar irradiation, temperature, and shading. It dynamically adjusts the operating parameters of the panels to extract the maximum available power. This optimization leads to improved energy harvest, especially in situations where solar conditions are less than ideal, such as cloudy or partially shaded environments. MPPT ensures that the system can capture as much energy as possible from the available sunlight.
3. **Faster Charging:** With its ability to optimize power output, MPPT charging systems can deliver higher charging currents to EV batteries. This results in faster charging times, allowing EV owners to recharge their vehicles more quickly. By maximizing power transfer, MPPT-based charging systems offer efficient and time-saving charging solutions.
4. **Adaptability to Variable Conditions:** Solar PV systems with MPPT can effectively handle variations in solar conditions. The algorithms constantly monitor and adjust the operating point of the panels, allowing the system to adapt to changes in solar irradiation, temperature, and shading. This flexibility ensures that the system operates optimally and provides consistent charging performance, regardless of environmental fluctuations.
5. **Optimal Utilization of Solar Energy:** MPPT-based charging systems maximize the utilization of available solar energy. By operating the solar panels at their peak power output, MPPT ensures that the energy generated is efficiently used for charging EV batteries. This not only reduces reliance on grid electricity but also promotes the use of clean, renewable energy sources, contributing to a greener and more sustainable energy ecosystem.
6. **Compatibility with Battery Systems:** MPPT charging systems can be integrated with various types of battery storage systems. This compatibility allows for efficient charging of EV batteries, as the charging current can be precisely controlled and optimized based on the battery's charging characteristics. MPPT ensures that the charging process is tailored to the specific requirements of the batteries, enhancing charging efficiency and battery lifespan.

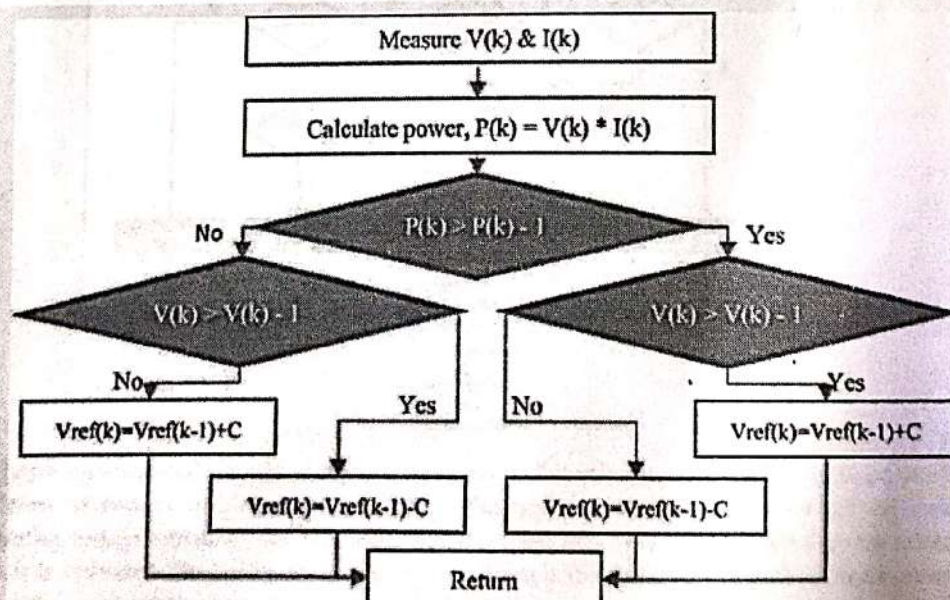
❖ Different MPPT algorithms:-

The maximum power supplied by the photovoltaic panels is not always stable and fixed in the same operating point; it varies with the weather conditions, such as solar irradiation, shadow, and temperature. To extract the maximum power, it is necessary to implement an MPPT algorithm that dynamically adjusts the extraction of the power. Convergence speed is one of the most important features among all different MPPT algorithms. Any improvement in the rise time of MPPT improves the reliability of the system and increases the power extraction and efficiency of the whole system.

1. Perturbation and Observation (P&O)

Perturb and observe algorithm is simple and does not require previous knowledge of the PV generator characteristics or the measurement of solar intensity and cell temperature and is easy to implement with analogue and digital circuits. It perturbs the operating point of the system causing the PV array terminal voltage to fluctuate around the MPP voltage even if the solar irradiance and the cell temperature are constants [6].

Moreover, it is the most widely used and workhorse MPPT algorithm because of its balance between performance and simplicity. However, it suffers from the lack of speed and adaptability which is necessary for tracking the fast transients under varying environmental conditions [7]. It is simple and straightforward technique but degraded performance is achieved due to the trade-off between accuracy and speed-upon selecting the step size [8]. Figure 4 shows the P&O algorithm diagram.



Incremental Conductance :-

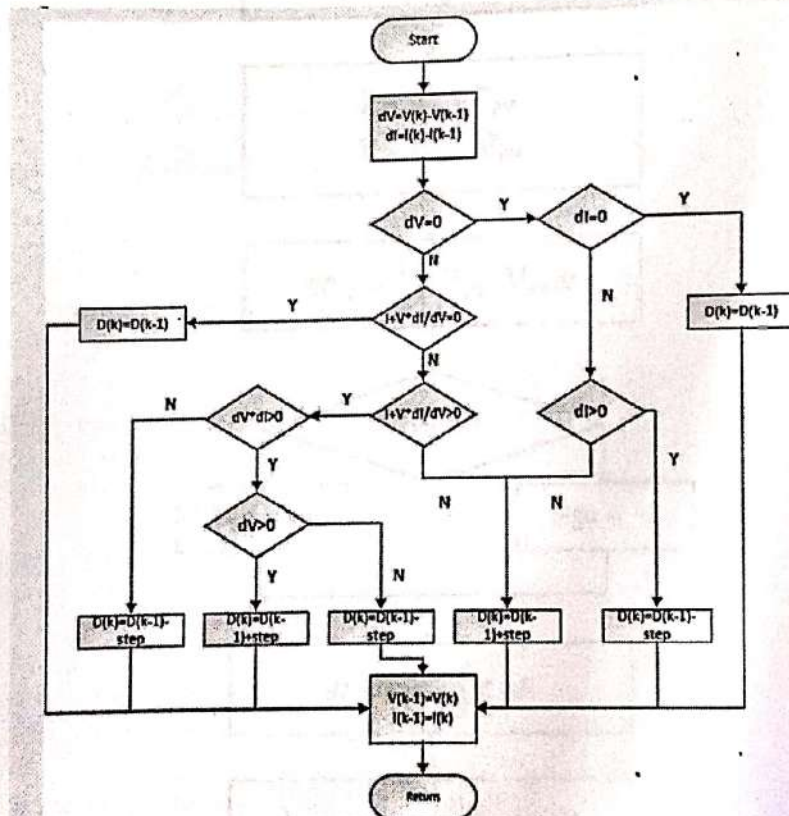
The incremental conductance method is based on the principle that the slope of the PV array power curve is zero at the MPP, so that, $\Delta P / \Delta V = 0$ with $P = VI$

Considering that

$$\begin{aligned} \frac{\Delta I}{\Delta V} &= -\frac{I}{V} \quad \text{if } P = \text{MPP}, \\ \frac{\Delta I}{\Delta V} &> -\frac{I}{V} \quad \text{if } P < \text{MPP}, \\ \frac{\Delta I}{\Delta V} &< -\frac{I}{V} \quad \text{if } P > \text{MPP}, \end{aligned} \quad (1)$$

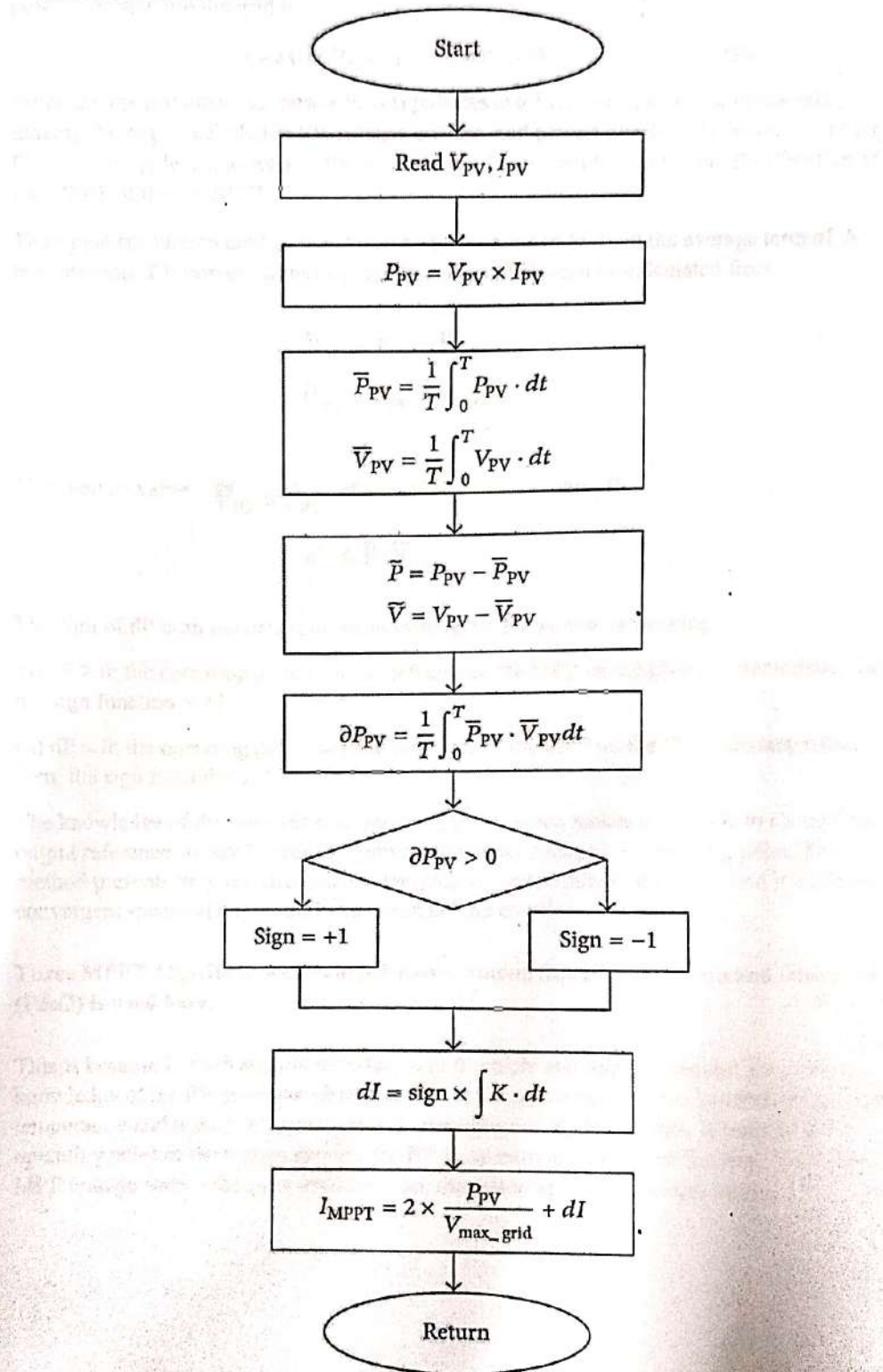
the MPP can be tracked by comparing the instantaneous conductance I/V with the incremental conductance $\Delta I / \Delta V$.

The algorithm increments or decrements the reference until the condition $\Delta I / \Delta V = -I/V$ is achieved. Once the maximum power is reached, the operation of the PV array is maintained at this point. It is an



effective algorithm and requires high sampling rates and fast calculations of the power slope. It offers different advantages which are good tracking efficiency and automatic adjustment of the module operating voltage with no oscillations. Also, the response is improved and the control for the extracted power is optimized. The implementation of this algorithm in the control unit is difficult and expensive, but with recent developments in microcontrollers it became more cost effective. It lends itself well to DSP control, which can easily keep track of previous values of voltage and current.

2. Ripple Correlation:-



If the grid current $I_{grid}(t)$ is in phase with the grid voltage $V_{grid}(t)$, the instantaneous value of power injected into the grid is

$$P_{grid}(t) = V_{grid} I_{grid} (1 + \cos 2\omega t) \quad (2)$$

From (2), the instantaneous power $P_{grid}(t)$ pulsates at a frequency twice that of the grid, making the ripple included in PV voltage, current, and power. For that, the behaviour of the PV voltage ripple V_{pv} as well as PV power ripple P_{pv} is used to find PV out the direction of the MPPT of the SP-NPCI.

The ripple correlation control uses the average function to find out the average term of the instantaneous PV power P_{pv} and voltage V_{pv} . P_{pv} and V_{pv} can be calculated from

$$\tilde{P}_{PV} = P_{PV} - \bar{P}_{PV},$$

$$\tilde{V}_{PV} = V_{PV} - \bar{V}_{PV}.$$

The average value $\bar{V}_{PV} \times \bar{P}_{PV}$ of function is the output of dP

$$\partial P = \overline{\tilde{P} \cdot \tilde{V}}.$$

The sign of dP is an indication of region where the PV system is working:

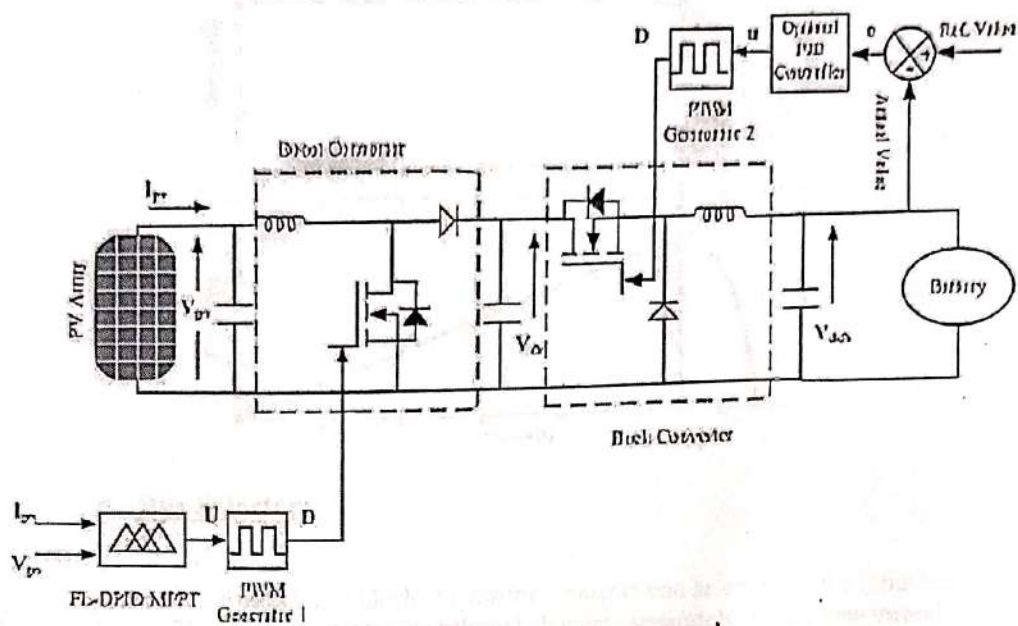
- (i) $dP > 0$: the operating point is on the left side of the MPP on the (P-V) characteristic; then, the sign function is +1.
- (ii) $dP < 0$: the operating point is on the right side of the MPP on the (P-V) characteristic; then, the sign function is -1.

The knowledge of the instantaneous operating point region makes it possible to change the output reference current in order to approach the maximum power operating point. This method presents very fast dynamics converging asymptotically to the MPP, and it achieves convergent speeds at a rate similar to the switching converter frequency.

Three MPPT Algorithm are discussed above. Among those Perturbation and Observation (P&O) is used here.

This is because Perturb and observe algorithm is simple and does not require previous knowledge of the PV generator characteristics or the measurement of solar intensity and cell temperature and is easy to implement with analogue and digital circuits. It perturbs the operating point of the system causing the PV array terminal voltage to fluctuate around the MPP voltage even if the solar irradiance and the cell temperature are constants.

❖ MPPT Circuit description

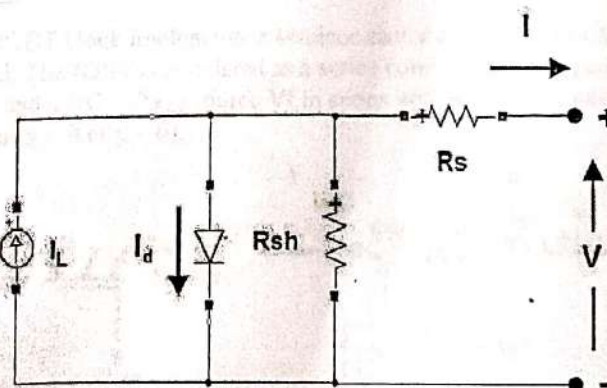


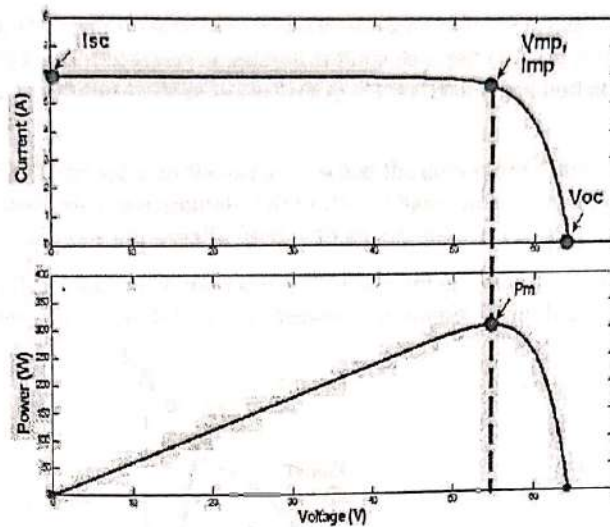
• Circuit Description-

• PV Array:

The PV Array block implements an array of photovoltaic (PV) modules. The array is built of strings of modules connected in parallel, each string consisting of modules connected in series.

The PV Array block is a five-parameter model using a light-generated current source (I_L), diode, series resistance (R_s), and shunt resistance (R_{sh}) to represent the irradiance- and temperature-dependent I-V characteristics of the modules.





- **Bus Selector:**

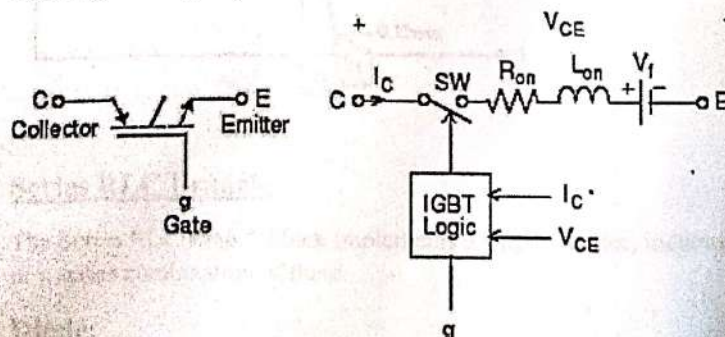
The Bus Selector block outputs the elements you select from the input bus. The block can output the selected elements separately or in a new virtual bus. The Bus Selector block does not support mixing message and signal elements as outputs.

- **Series RLC Load:**

The Series RLC Load block implements a linear load as a series combination of R L C elements. At the specified frequency, the load exhibits a constant impedance. The active and reactive powers absorbed by the load are proportional to the square of the applied voltage.

- **IGBT:**

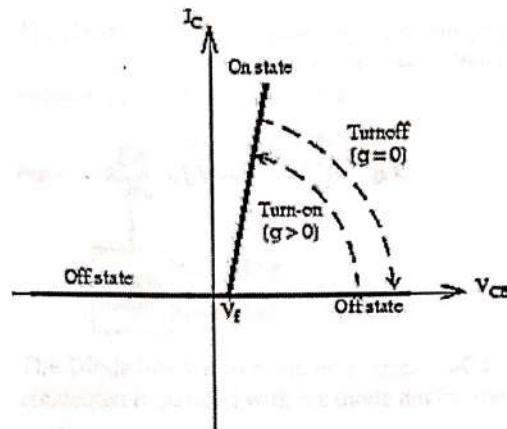
The IGBT block implements a semiconductor device controllable by the gate signal. The IGBT is simulated as a series combination of a resistor R_{on} , inductor L_{on} , and a DC voltage source V_f in series with a switch controlled by a logical signal ($g > 0$ or $g = 0$).



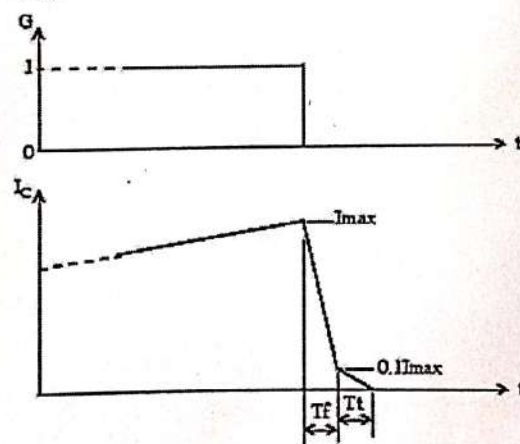
The IGBT turns on when the collector-emitter voltage is positive and greater than V_f and a positive signal is applied at the gate input ($g > 0$). It turns off when the collector-emitter voltage is positive and a 0 signal is applied at the gate input ($g = 0$).

The IGBT device is in the off state when the collector-emitter voltage is negative. Note that many commercial IGBTs do not have the reverse blocking capability. Therefore, they are usually used with an antiparallel diode.

The IGBT block contains a series Rs-Cs snubber circuit, which is connected in parallel with the IGBT device (between terminals C and E).



The turnoff characteristic of the IGBT model is approximated by two segments. When the gate signal falls to 0, the collector current decreases from I_{max} to $0.1 I_{max}$ during the fall time (T_f), and then from $0.1 I_{max}$ to 0 during the tail time (T_t).

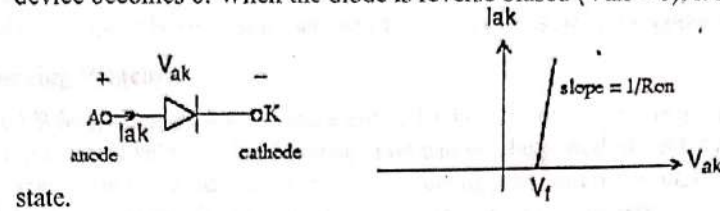


- **Series RLC Branch:**

The Series RLC Branch block implements a single resistor, inductor, or capacitor, or a series combination of these.

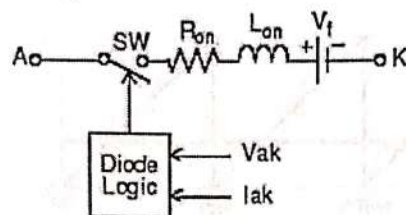
- **Diode:**

The diode is a semiconductor device that is controlled by its own voltage V_{ak} and current I_{ak} . When a diode is forward biased ($V_{ak} > 0$), it starts to conduct with a small forward voltage V_f across it. It turns off when the current flow into the device becomes 0. When the diode is reverse biased ($V_{ak} < 0$), it stays in the off



state.

The Diode block is simulated by a resistor, an inductor, and a DC voltage source connected in series with a switch. The switch operation is controlled by the voltage V_{ak} and the current I_{ak} .



The Diode block also contains a series Rs-Cs snubber circuit that can be connected in parallel with the diode device (between nodes A and K).

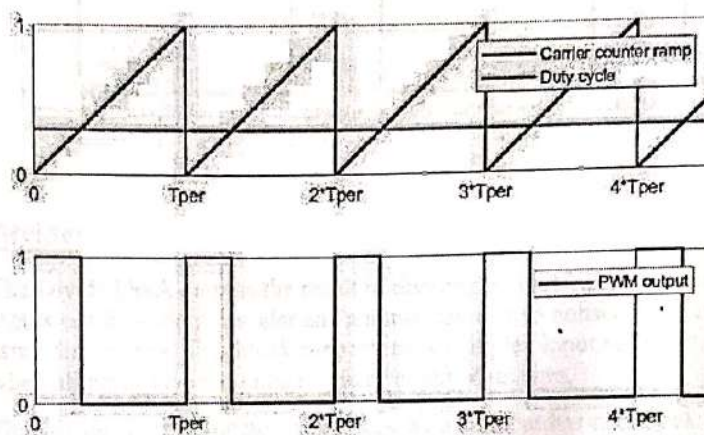
- **PWM Generator:**

The PWM Generator block implements a PWM generator. The pulse width modulation technique controls power transfer from one electrical component to another by quickly switching between full power transfer and no power transfer.

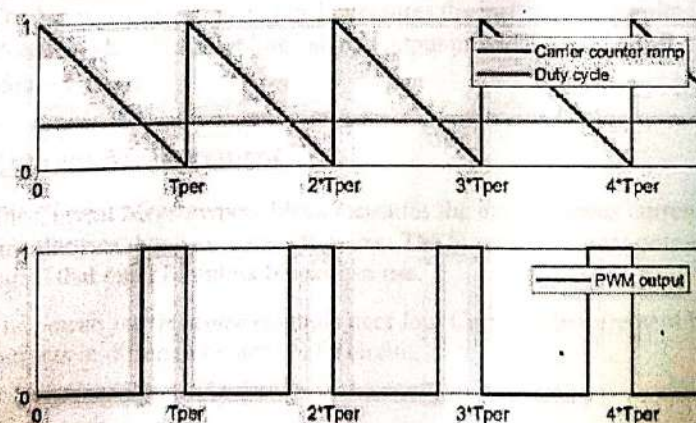
Working Principle

The PWM generator block outputs either 1 when the duty cycle is greater than the carrier counter value, or 0 otherwise. You can set the period of each cycle by specifying the timer period T_{per} . You can change the initial output, or phase, of the PWM output by specifying one of three types of carrier counters:

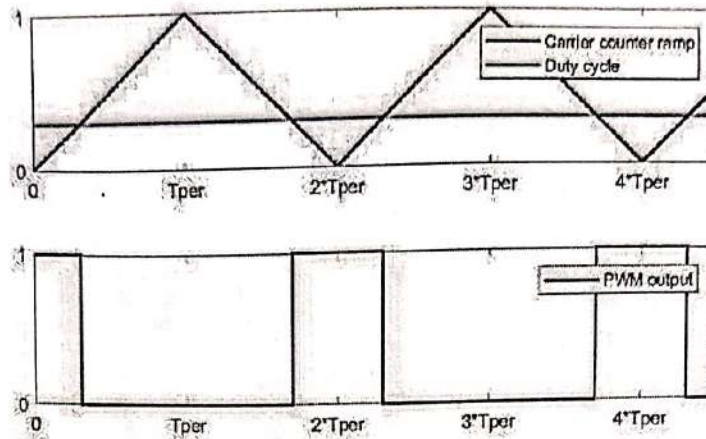
- Up counter — The PWM output signal initializes at the start of the on cycle. This graphic shows the carrier counter signal and the corresponding PWM output.



- Down counter — The PWM output signal initializes at the start of the off cycle. This graphic shows the carrier counter signal and the corresponding PWM output.



- **Up-down counter** — The PWM output signal initializes halfway through the on cycle. This graphic shows the carrier counter signal and the corresponding PWM output.



- **Divide:**

The Divide block outputs the result of dividing its first input by its second. The inputs can be scalars, a scalar and a nonscalar, or two nonscalars that have the same dimensions. This block supports only complex input values at division ports when all ports have the same single or double data type.

The Divide block is functionally a Product block that has two block parameter values preset:

- **Multiplication** — Element-wise(./)
- **Number of Inputs** — */

- **Voltage Measurement:**

The Voltage Measurement block measures the instantaneous voltage between two electric nodes. The Simulink® signal output provides a signal that other Simulink blocks can use.

- **Current Measurement:**

The Current Measurement block measures the instantaneous current flowing in any electrical block or connection line. The Simulink® signal output provides a signal that other Simulink blocks can use.

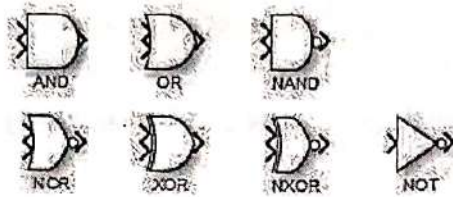
The *power_currmeasure* example uses four Current Measurement blocks to read currents in different branches of a circuit.

- **Logical Operator:**

The Logical Operator block performs the specified logical operation on its inputs. An input value is true (1) if it is nonzero and false (0) if it is zero.

You select the Boolean operation connecting the inputs with the **Operator** parameter list. If you select rectangular as the **Icon shape** property, the name of the selected operator displays on the block icon. If you select distinctive as the **Icon shape** property, the name of the selected operator does not display on the block icon. This table shows supported operations:

If you select distinctive as the **Icon shape**, the block appearance indicates its function. Simulink® software displays a distinctive shape for the selected operator, conforming to the IEEE® Standard Graphic Symbols for Logic Functions.



To specify the number of input ports, use the **Number of input ports** parameter. The output type is specified using the **Output data type** parameter. An output value is 1 if TRUE and 0 if FALSE.

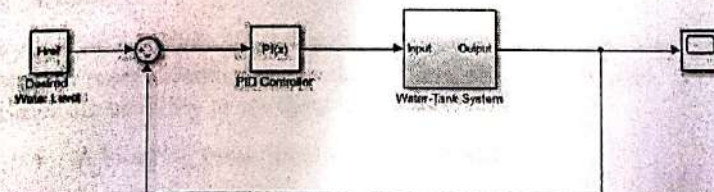
- **Battery:**

Implements a generic battery model for most popular battery types. Temperature and aging (due to cycling) effects can be specified for Lithium-Ion battery type.

- **Discrete PID Controller:**

The Discrete PID Controller block implements a PID controller (PID, PI, PD, P only, or I only). The block is identical to the PID Controller block with the **Time domain** parameter set to Discrete-time.

The block output is a weighted sum of the input signal, the integral of the input signal, and the derivative of the input signal. The weights are the proportional, integral, and derivative gain parameters. A first-order pole filters the derivative action.

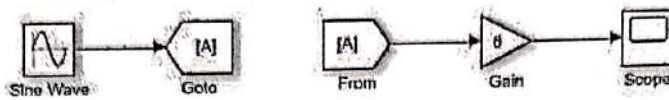


- **Go:**

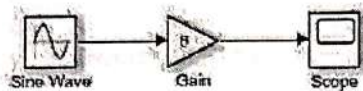
The goto block passes its input to its corresponding From blocks. The input can be a real- or complex-valued signal or vector of any data type. From and goto blocks allow you to pass a signal from one block to another without actually connecting them.

A goto block can pass its input signal to more than one From block, although a From block can receive a signal from only one goto block. The input to that goto block is passed to the From blocks associated with it as though the blocks were physically connected.

For example, this model uses a goto block and a From block.



An equivalent model passes the Sine Wave block signal directly to the Gain block.



goto blocks and From blocks are matched by the use of goto tags.

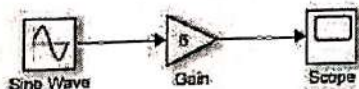
• **From:**

The from block accepts a signal from a corresponding Goto block, then passes it as output. The data type of the output is the same as that of the input from the Goto block. from and Goto blocks allow you to pass a signal from one block to another without actually connecting them.

For example, this model uses a Goto block and a from block.



An equivalent model passes the Sine Wave block signal directly to the Gain block.



A from block can receive its signal from only one Goto block, although a Goto block can pass its signal to more than one from block.

To associate a Goto block with a from block, enter the Goto block tag in the **Goto Tag** parameter.

The visibility of a Goto block tag determines the from blocks that can receive its signal. For more information, see Goto and Goto Tag Visibility. The block indicates the visibility of the Goto block tag:

- A local tag name is enclosed in brackets ([]).
- A scoped tag name is enclosed in braces ({}).
- A global tag name appears without additional characters.

The from block supports signal label propagation.

MPPT Control Code-

```
function duty = MPPT_Control(V, I,deltaD_in)

duty_init = 0.05;
duty_min = 0;
duty_max = 0.75;

persistent Vold Pold duty_old;

if isempty(Vold)
    Vold=0;
    Pold=0;
    duty_old=duty_init;
end
P= V*I;
dV= V - Vold;
dP= P - Pold;
duty = duty_old;
deltaD=deltaD_in;

if dP ~= 0
    if dP < 0
        if dV < 0
            duty = duty_old - deltaD;
        else
            duty = duty_old + deltaD;
        end
    else
        if dV < 0
            duty = duty_old + deltaD;
        else
            duty = duty_old - deltaD;
        end
    end
end

if duty >= duty_max
    duty=duty_max;
elseif duty<duty_min
    duty=duty_min;
end

duty_old=duty;
Vold=V;
Pold=P;
```


Control Code Description-

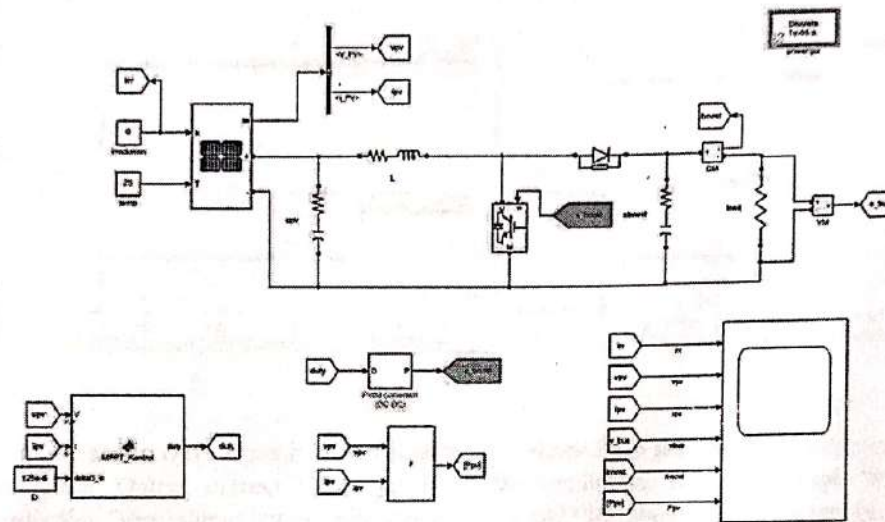
The provided code is an implementation of the MPPT (Maximum Power Point Tracking) control algorithm. It is used to optimize the power output of a photovoltaic system by dynamically adjusting the duty cycle of a power converter.

Let's go through the code step by step:

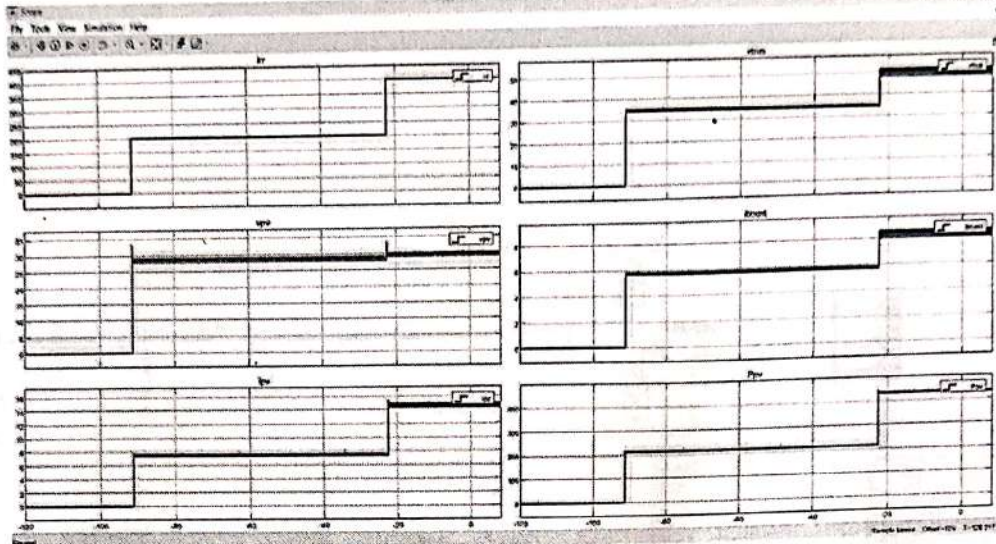
- [1] The function `MPPT_Control` takes three input arguments: `V` (voltage), `I` (current), and `deltaD_in` (a parameter used to adjust the duty cycle).
- [2] The initial duty cycle value is set to 0.05, and the minimum and maximum allowable duty cycle values are defined as 0 and 0.75, respectively.
- [3] The variables `Vold`, `Pold`, and `duty_old` are declared as persistent variables. Persistent variables retain their values between function calls. This allows the function to remember the previous values of voltage, power, and duty cycle.
- [4] Inside the function, the first time the function is called, the persistent variables are initialized to zero and the initial duty cycle value.
- [5] The power (`P`) is calculated by multiplying the voltage (`V`) and current (`I`).
- [6] The changes in voltage (`dV`) and power (`dP`) are calculated by subtracting the previous values (`Vold` and `Pold`) from the current values (`V` and `P`).
- [7] The current duty cycle (`duty`) is assigned the value of the previous duty cycle (`duty_old`).
- [8] The desired change in duty cycle (`deltaD`) is assigned the value of the input parameter `deltaD_in`.
- [9] The code then checks if there is a change in power (`dP`). If `dP` is not equal to zero, it means there has been a change in power, and the algorithm proceeds to adjust the duty cycle.
- [10] If `dP` is negative, the code checks if `dV` is also negative. If both `dP` and `dV` are negative, it suggests that the power is decreasing and the system is moving away from the maximum power point. In this case, the duty cycle is decreased by `deltaD` to bring the system closer to the maximum power point. Otherwise, if `dV` is positive, the duty cycle is increased by `deltaD`.
- [11] If `dP` is positive, the code checks if `dV` is negative. If `dP` is positive and `dV` is negative, it suggests that the power is increasing and the system is moving towards the maximum power point. In this case, the duty cycle is increased by `deltaD` to bring the system closer to the maximum power point. Otherwise, if `dV` is positive, the duty cycle is decreased by `deltaD`.
- [12] After adjusting the duty cycle, the code checks if the duty cycle exceeds the maximum duty cycle (`duty_max`) or falls below the minimum duty cycle (`duty_min`). If `duty` is greater than or equal to `duty_max`, it is set to `duty_max`. If `duty` is less than `duty_min`, it is set to `duty_min`.
- [13] Finally, the value of the current duty cycle (`duty`) is stored in the persistent variable `duty_old`, and the current values of voltage (`V`) and power (`P`) are stored in the persistent variables `Vold` and `Pold`, respectively. The code implements a basic MPPT control algorithm that continuously adjusts the duty cycle based on the changes in power and voltage. The aim is to track the maximum power point of the photovoltaic system and optimize its power output.

Simulation

MPPT Simulation in MATLAB:-

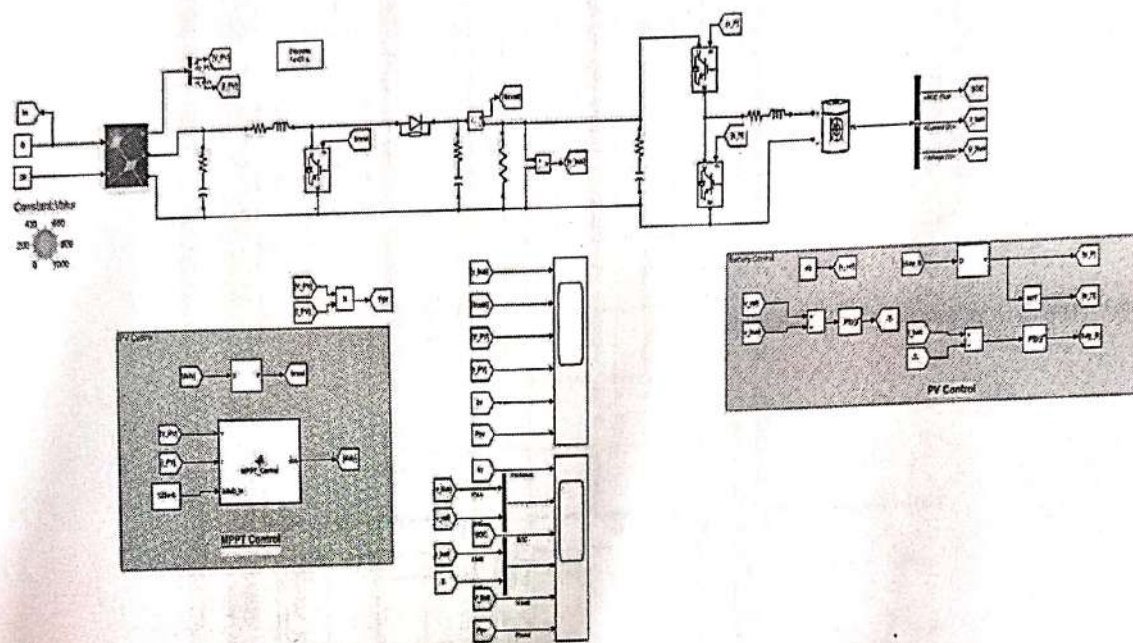


- Simulation result of MPPT circuit

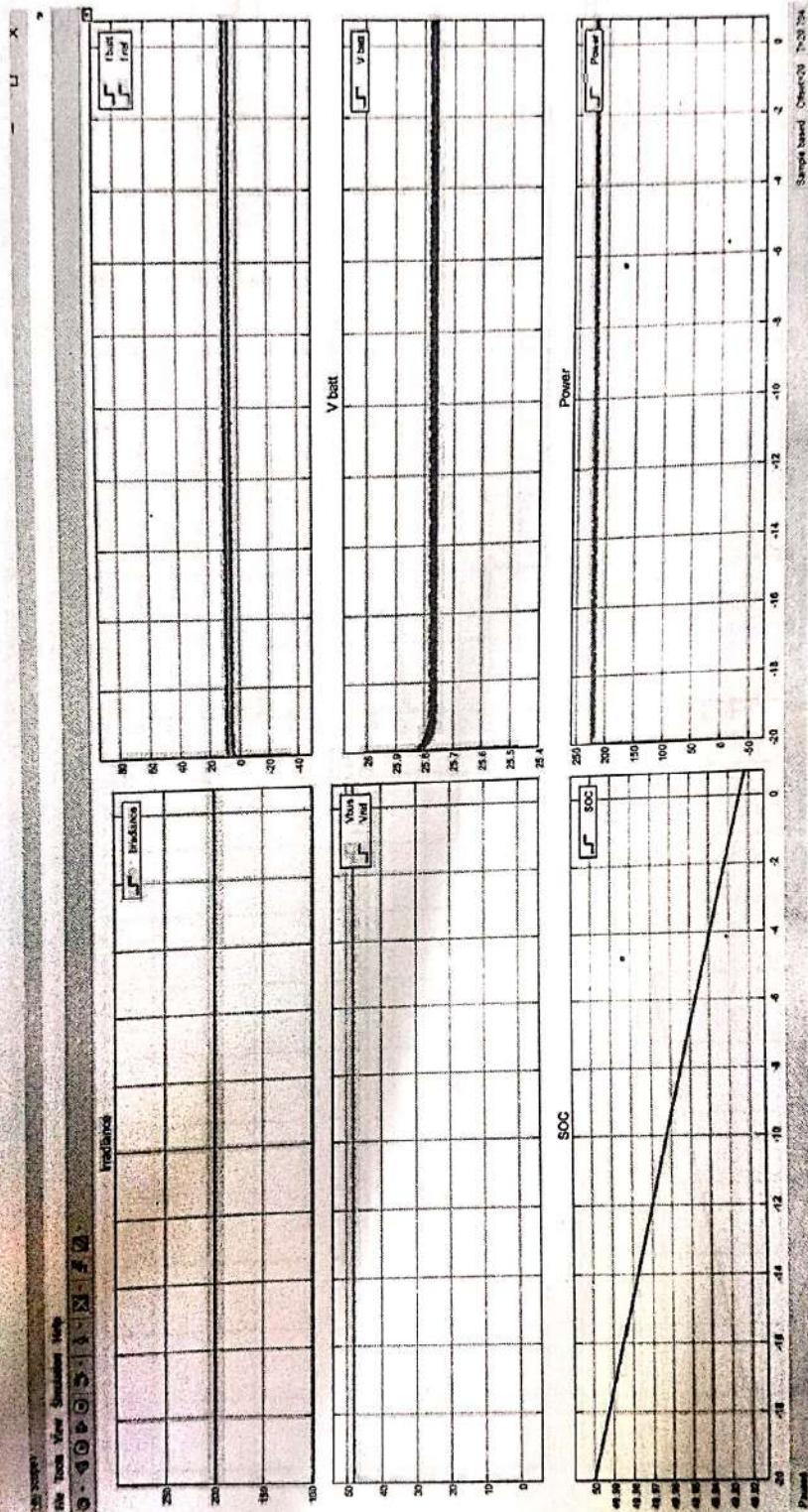


From the above waveforms it is clear that output voltage from the PV array is depends on the irradiation. Output current also depends on the irradiation. In above graph 'irr' shows irradiation, 'Vpv' shows output voltage of PV array and 'ipv' shows output current of PV array. When irradiation is 0, there is no current or output voltage in PV array. With increase in irradiation level output voltage and output current increases from the PV array.

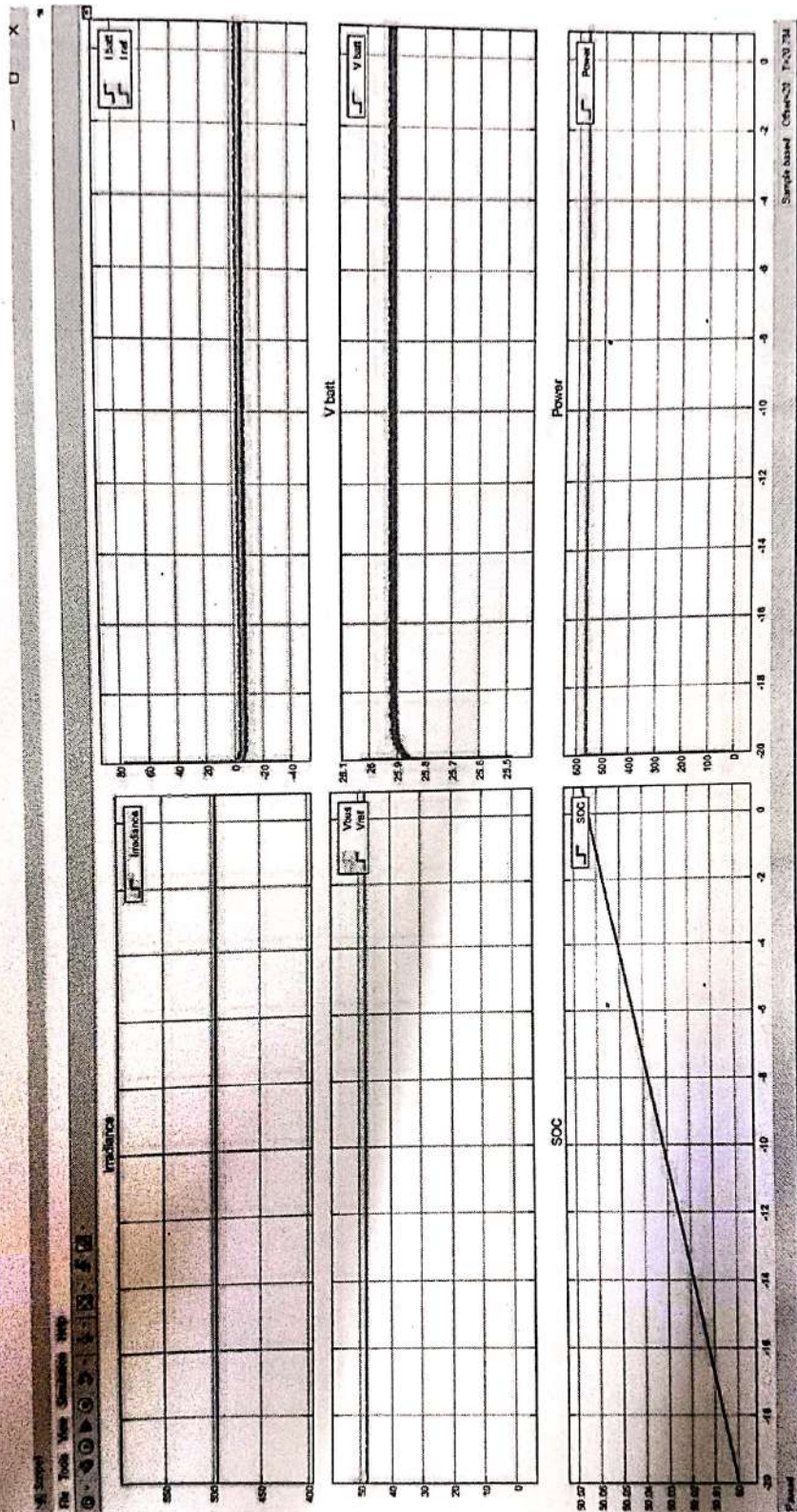
MPPT With Charging Circuit In MATLAB:-



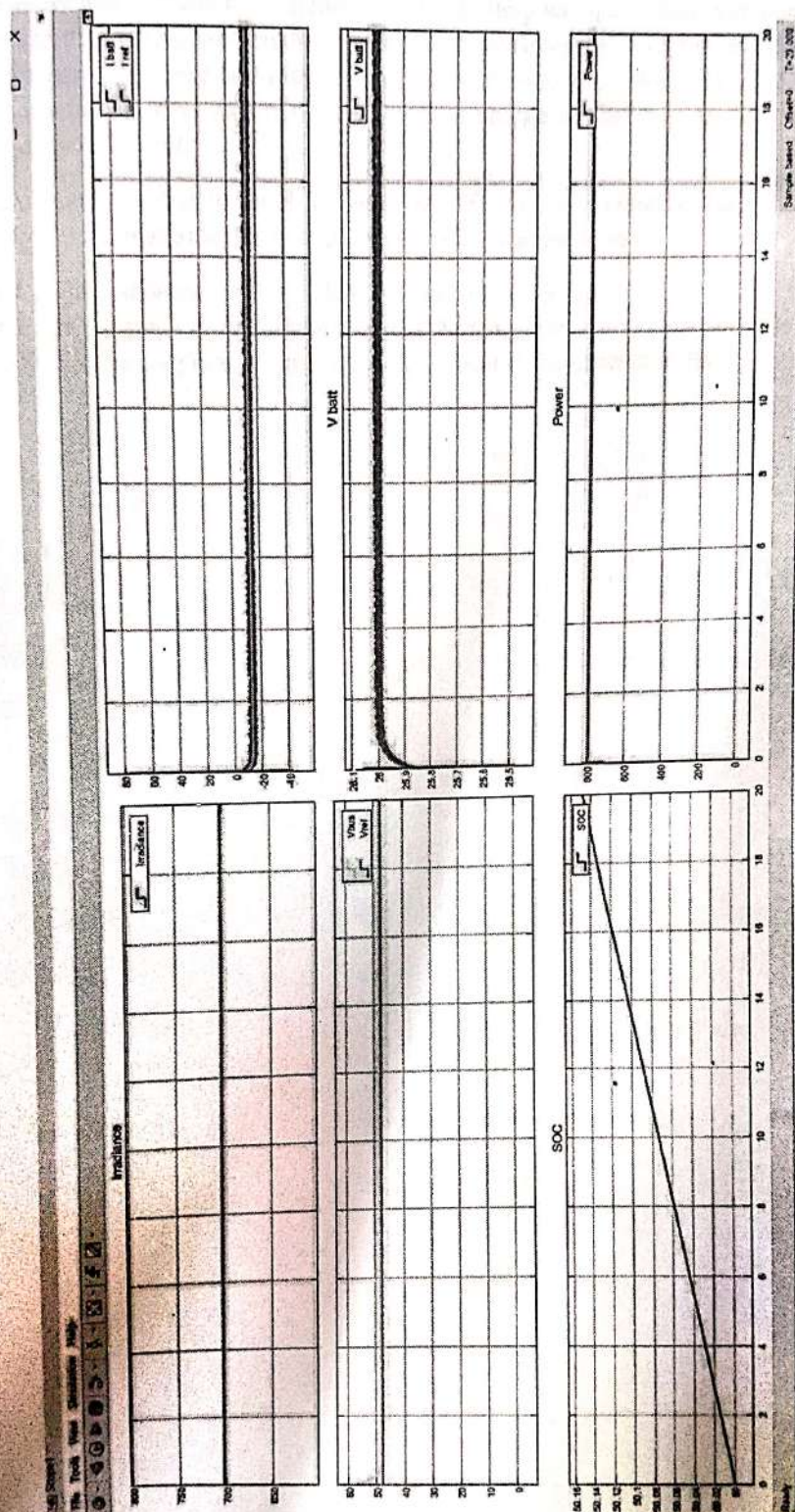
- Simulation Result-
Irradiation 200



Irradiation 500



Irradiation 700



- **Analysis-**

From the above graph shows the variation of SOC, Battery voltage (V batt) and battery current due to the variation of different irradiation level. Battery voltage Initially fluctuates but remains constant for particular irradiation level. When irradiation level is 200, SOC decreasing over time. i.e., irradiation level 200 is not sufficient to charge the battery. At this level battery voltage is about 25.8 volt.

In the next graph irradiation level increases to 500. At this irradiation level battery starts charging from its initial SOC level of 50. Here battery voltage is about 25.9 volt.

In the last graph irradiation level is 700, battery voltage is about 26 volts and battery start charging. From the graph it is clearly visible that charging rate is more than previous irradiation level of 500. So charging rate of battery is proportional to the irradiation level.